* 0000800000001 *

Cambridge Assessment International Education

# Cambridge International AS & A Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*6441974659*

**MATHEMATICS** **9709/12**

Paper 1 Pure Mathematics 1 **October/November 2024**

**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

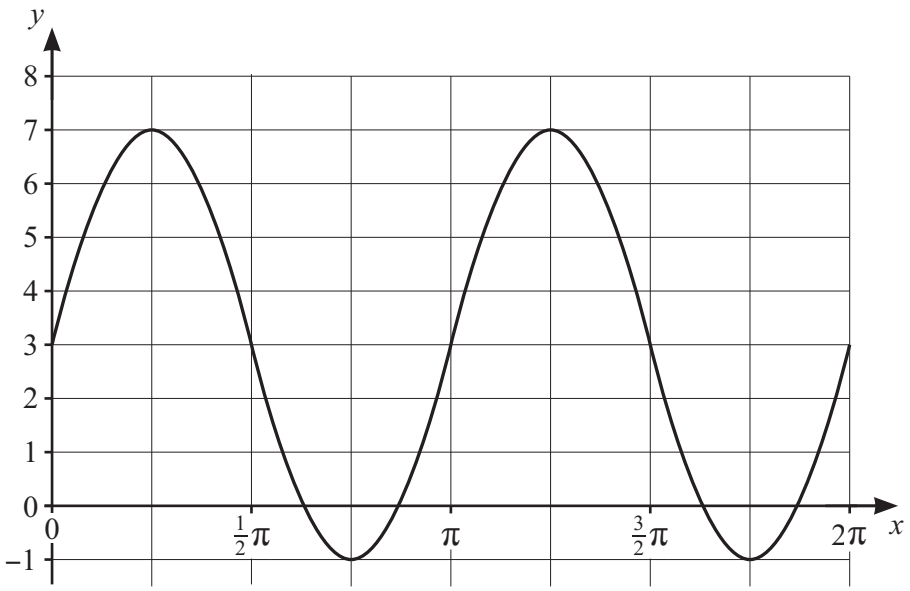
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

DC (CE/CB) 337384/5
© UCLES 2024

**[Turn over**

**1**




The diagram shows the curve with equation $y = a\sin(bx) + c$ for $0 \leqslant x \leqslant 2\pi$, where $a$, $b$ and $c$ are positive constants.

**(a)** State the values of $a$, $b$ and $c$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

**(b)** For these values of $a$, $b$ and $c$, determine the number of solutions in the interval $0 \leqslant x \leqslant 2\pi$ for each of the following equations:

**(i)** $a\sin(bx) + c = 7 - x$ [1]

.....

.....

**(ii)** $a\sin(bx) + c = 2\pi(x - 1)$. [1]

.....

.....

DO NOT WRITE IN THIS MARGIN

**2** The first term of an arithmetic progression is $-20$ and the common difference is 5.

**(a)** Find the sum of the first 20 terms of the progression. [2]

.....

.....

.....

.....

.....

.....

.....

It is given that the sum of the first $2k$ terms is 10 times the sum of the first $k$ terms.

**(b)** Find the value of $k$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**3** The equation of a curve is $y = 2x^2 - 3$. Two points $A$ and $B$ with $x$-coordinates 2 and $(2 + h)$ respectively lie on the curve.

**(a)** Find and simplify an expression for the gradient of the chord $AB$ in terms of $h$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**(b)** Explain how the gradient of the curve at the point $A$ can be deduced from the answer to part **(a)**, and state the value of this gradient. [2]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

**4** Find the term independent of $x$ in the expansion of each of the following:

**(a)** $\left(x+\frac{3}{x^2}\right)^6$ [2]

.....

.....

.....

.....

.....

.....

.....

**(b)** $(4x^3-5)\left(x+\frac{3}{x^2}\right)^6$. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**5** The function f is defined by $\mathrm{f}(x) = \dfrac{2x+1}{2x-1}$ for $x < \dfrac{1}{2}$.

**(a) (i)** State the value of $\mathrm{f}(-1)$. [1]

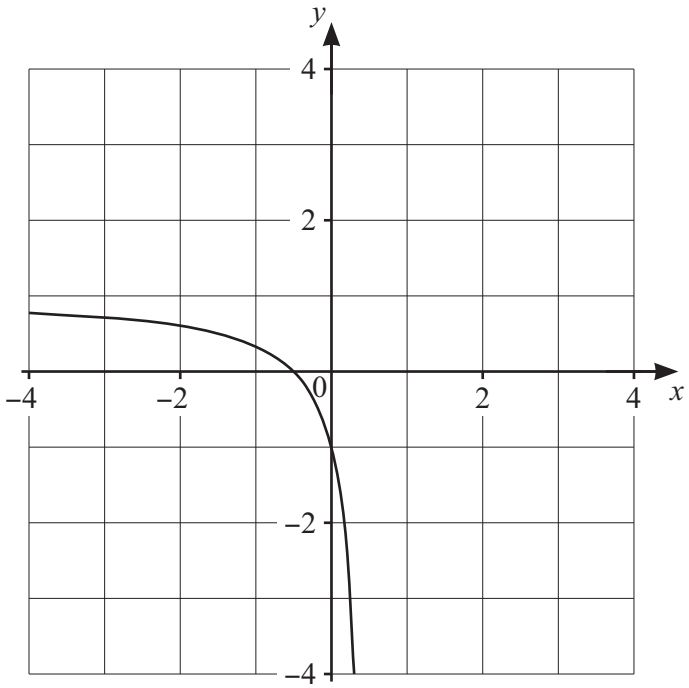
.....

.....

.....

.....

**(ii)**




The diagram shows the graph of $y = \mathrm{f}(x)$. Sketch the graph of $y = \mathrm{f}^{-1}(x)$ on this diagram. Show any relevant mirror line. [2]

**(iii)** Find an expression for $\mathrm{f}^{-1}(x)$ and state the domain of the function $\mathrm{f}^{-1}$. [4]

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

The function g is defined by $g(x) = 3x + 2$ for $x \in \mathbb{R}$.

**(b)** Solve the equation $f(x) = gf\left(\frac{1}{4}\right)$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

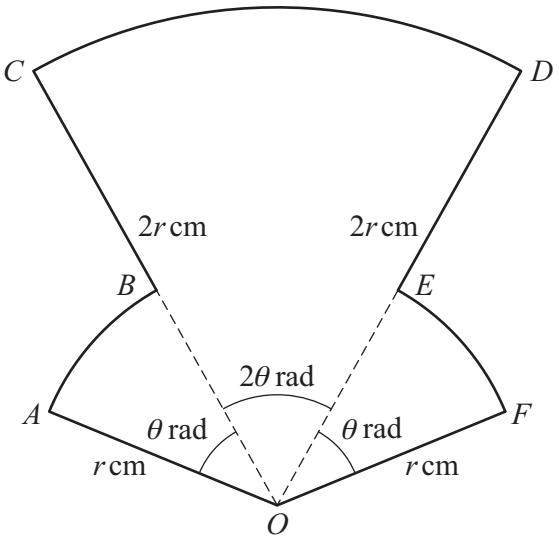
.....

.....

.....

.....

6




The diagram shows a metal plate $OABCDEF$ consisting of sectors of two circles, each with centre $O$. The radii of sectors $AOB$ and $EOF$ are $r$ cm and the radius of sector $COD$ is $2r$ cm.
Angle $AOB$ = angle $EOF = \theta$ radians and angle $COD = 2\theta$ radians.

It is given that the perimeter of the plate is 14 cm and the area of the plate is $10\,\text{cm}^2$.

Given that $r > \frac{3}{2}$ and $\theta < \frac{3}{4}$, find the values of $r$ and $\theta$. [6]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**7** **(a)** By expressing $-2x^2+8x+11$ in the form $-a(x-b)^2+c$, where $a$, $b$ and $c$ are positive integers, find the coordinates of the vertex of the graph with equation $y=-2x^2+8x+11$. [3]

.....

.....

.....

.....

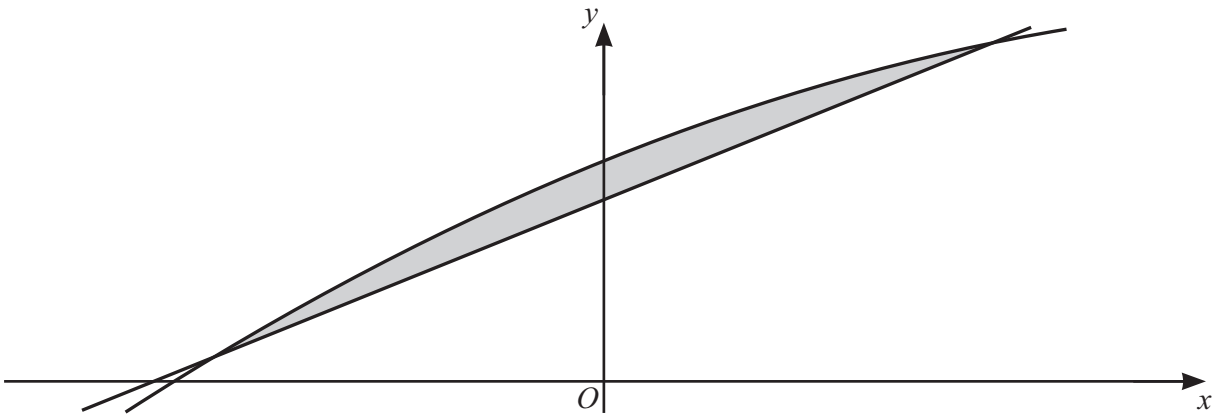
.....

.....

.....

.....

**(b)**




The diagram shows part of the curve with equation $y=-2x^2+8x+11$ and the line with equation $y=8x+9$.

Find the area of the shaded region. [5]

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**8** The equation of a circle is $x^2 + y^2 + px + 2y + q = 0$, where $p$ and $q$ are constants.

**(a)** Express the equation in the form $(x-a)^2 + (y-b)^2 = r^2$, where $a$ is to be given in terms of $p$ and $r^2$ is to be given in terms of $p$ and $q$. [2]

.....

.....

.....

.....

.....

.....

.....

The line with equation $x + 2y = 10$ is the tangent to the circle at the point $A(4, 3)$.

**(b) (i)** Find the equation of the normal to the circle at the point $A$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

**(ii)** Find the values of $p$ and $q$. [5]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**9** The equation of a curve is $y = \frac{1}{2}k^2x^2 - 2kx + 2$ and the equation of a line is $y = kx + p$, where $k$ and $p$ are constants with $0 < k < 1$.

**(a)** It is given that one of the points of intersection of the curve and the line has coordinates $\left(\frac{5}{2}, \frac{1}{2}\right)$.

Find the values of $k$ and $p$, and find the coordinates of the other point of intersection. [7]

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

.....

.....

.....

.....

.....

.....

.....

.....

**(b)** It is given instead that the line and the curve do **not** intersect.

Find the set of possible values of $p$. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**10** A function f with domain $x > 0$ is such that $f'(x) = 8(2x-3)^{\frac{1}{3}} - 10x^{\frac{2}{3}}$. It is given that the curve with equation $y = f(x)$ passes through the point (1, 0).

**(a)** Find the equation of the normal to the curve at the point (1, 0). [3]

.....

.....

.....

.....

.....

**(b)** Find $f(x)$. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

DO NOT WRITE IN THIS MARGIN

It is given that the equation $f'(x) = 0$ can be expressed in the form

$$125x^2 - 128x + 192 = 0.$$

**(c)** Determine, making your reasoning clear, whether f is an increasing function, a decreasing function or neither. [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**Additional page**

If you use the following page to complete the answer to any question, the question number must be clearly shown.

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**BLANK PAGE**

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

**BLANK PAGE**

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.